

Invention Disclosure Format (IDF)-B	Document No.	02-IPR-R003
	Issue No/Date	2/01.02.2024
	Amd. No/Date	0/00.00.0000

1. Title of the invention:

A Computer-Implemented System and Method for Asynchronous Multi-Attribute State Propagation and Temporal Dynamic Envelope Extraction Across Graph Data Structures

2. Field / Area of invention:

Attribute	Details & Technical Classification (Strictly G06F)
Primary Technical Field	Computer Systems, Graph Computing Data Structures, and Asynchronous Event-Driven Digital Data Processing
Core Computational Domains	Asynchronous Directed Graph Traversal, Cryptographic State-Transition Verification, Multi-Attribute Vector Scaling, Dynamic State Deflection Envelope Detection
Cross-Domain Embodiments	Packet Routing & Telecom Mesh Networks, Distributed Cloud Microservice Pipelines, Enterprise Resource & Operational Telemetry
Technical Architecture	In-Memory Temporal Multigraph Database Core, Chronological Event Queue Schedulers, Contiguous Adjacency Buffers, SHA-256 State Anchoring Engines
Statutory IPC Classifications	G06F 16/90 (Information Retrieval; Graph Data Structures & Traversal Indexes) G06F 9/48 (Program Execution Management; Asynchronous Event Dispatching & Queuing) G06F 17/16 (Vector & Matrix Computational Operations) G06F 21/64 (Data Integrity Verification; Cryptographic State Anchoring)

Statutory Classification & Examination Note: The claimed invention fundamentally pertains to the technical domain of computer-implemented digital data processing, graph data structures, asynchronous event-queue scheduling, and cryptographic integrity verification under IPC Class G06F. In accordance with Indian Patent Office CRI Guidelines (2017), the invention solves technical problems internal to computing systems, and all business methods are explicitly excluded from the claims.

3. Prior Patents and Publications from literature

3.1 Prior Art Survey

Sl. No.	Reference / Publication	Year	Key Contribution	Limitation / Technical Gap under Section 3(k)
1	US20240362563A1 'Financial risk graph system'	2024	Propagates dimensionless risk scores across entity networks using static weighted edges.	Abstract business scoring ; lacks coupled dual-transform vector operations, event queues, and temporal schedule shifting.
2	US12380389B2 'Entity-level financial risk graph'	2025	Entity-relationship mapping for corporate vulnerability scoring.	High-level entity mapping only ; lacks typed computational node registers and time-delayed state trajectory outputs.
3	US20260120033A1 'Business dependency graph'	2026	Quantifies systemic operational dependencies across organizations.	Lacks structured perturbation vector objects, dual-transform edges, and dynamic state deflection envelope extraction.
4	US10679166B2 'Supply chain financing system'	2020	Digital ledger managing trade financing instruments.	Administrative ledger only; does not simulate operational disturbances or extract dynamic state deflection envelopes.
5	US6167385A 'Event-driven logistics simulation'	2000	Event-driven simulation for physical supply chain inventory queues.	Restricted to discrete item counts; lacks dual-transform vector scaling, asynchronous graph traversal, and state convergence.

3.2 Key Academic Publications

Publication / Citation	Focus Area	Key Gap vs. Present Invention
Tangsucheeva & Prabhu, <i>Int. J. Prod. Econ.</i> (2013)	Analytical formulation of Cash-Flow Bullwhip Effect in supply networks.	Theoretical study only; lacks computational graph data structures, perturbation vectors, and simulation pipeline.
Temporal Attentive Graph Networks (TAGN), Preprints (2024)	Graph neural networks for financial contagion and anomaly detection.	Statistical ML classification; non-deterministic, lacks coupled dual-transform rules, and outputs no dynamic state envelope.

3.3 Identified Technical Gap in Computing Systems

No existing computational system unifies all foundational technical capabilities: (1) an in-memory temporal multigraph data structure of discrete computational nodes connected by dual-transform directed edges, (2) treatment of scalar perturbations as first-class typed state vector objects carrying numerical amplitudes and temporal metadata, (3) execution of simultaneous magnitude scaling and chronological schedule shifting in $O(|V| + |E|)$ single-pass linear time, (4) deterministic multi-path convergence resolution under numerical conservation, (5) cryptographic state-anchoring (SHA-256) ensuring bitwise execution invariance, and (6) automated extraction of a multi-parameter dynamic state deflection envelope. The present invention solves these technical data-processing bottlenecks.

4. Summary and background of the invention

4.1 Background & Technical Shortcomings of Prior Computing Architectures

Modern distributed networks (telecom mesh topologies, microservice task pipelines, and enterprise systems) exhibit complex multi-hop cascades. Conventional simulation tools fail because they lack technical mechanisms to model asynchronous causal propagation across cyclic multigraphs, producing exponential $O(V \cdot E^k)$ combinatorial path explosion, CPU cycle exhaustion, memory cache thrashing, and non-deterministic race conditions.

4.2 Summary of the Invention

The present invention is a universal technical computing system and computer-implemented method that processes operational perturbations through an in-memory temporal multigraph data structure $G = (V, E, T)$, propagating scalar amplitudes and temporal schedules across dual-transform edges in a single pass, resolving multi-path convergence at terminal accumulator nodes, and extracting an explicit **Dynamic State Deflection Envelope** data structure.

4.3 Internal Hardware Technical Effects (Section 3(k) Statutory Compliance)

In compliance with Section 3(k) of the Indian Patents Act, 1970, the CRI Guidelines (2017), and binding precedents of the Hon'ble Delhi High Court in *Ferid Allani v. Union of India* (2019) 265 DLT 420 and *Microsoft Technology Licensing LLC v. Assistant Controller of Patents & Designs* (2023) SCC OnLine Del 2772, the invention provides concrete technical solutions to technical problems internal to the operation of a computer:

- **Elimination of Graph Combinatorial Explosion:** Standard cyclic temporal multigraph traversal causes infinite loops or exponential $O(V \cdot E^k)$ path explosion. The present invention's single-pass coupled calculation ($\Delta A_v = \Delta A_u \cdot \omega_e \cdot \psi_e$ and $t_v = t_u + \delta_e$) utilizing a priority queue buffer guarantees $O(|V| + |E|)$ linear execution time, dramatically reducing CPU cycle consumption and RAM footprint.
- **Hardware Memory Cache Locality:** The temporal multigraph is indexed in memory using contiguous adjacency array buffers, maximizing L1/L2 hardware CPU cache hits and eliminating memory bus thrashing during multi-hop graph traversal.
- **Race-Condition-Free Parallel Execution:** The deterministic chronological event queue buffer enables multi-threaded worker cores to process asynchronous event streams without deadlocks or thread synchronization latency, achieving bitwise identical reproducibility.
- **Cryptographic State-Transition Integrity:** Pre- and post-simulation state arrays are anchored via SHA-256 cryptographic hash digests, ensuring tamper-evident data integrity across parallel computing nodes.

The claimed invention is therefore not a business method, mathematical method, algorithm, or computer programme per se, but an advanced digital data processing architecture.

FIG. 1: Novelty Differentiation Radar - Quantitative Prior Art Benchmarking (Scale: 1-10)

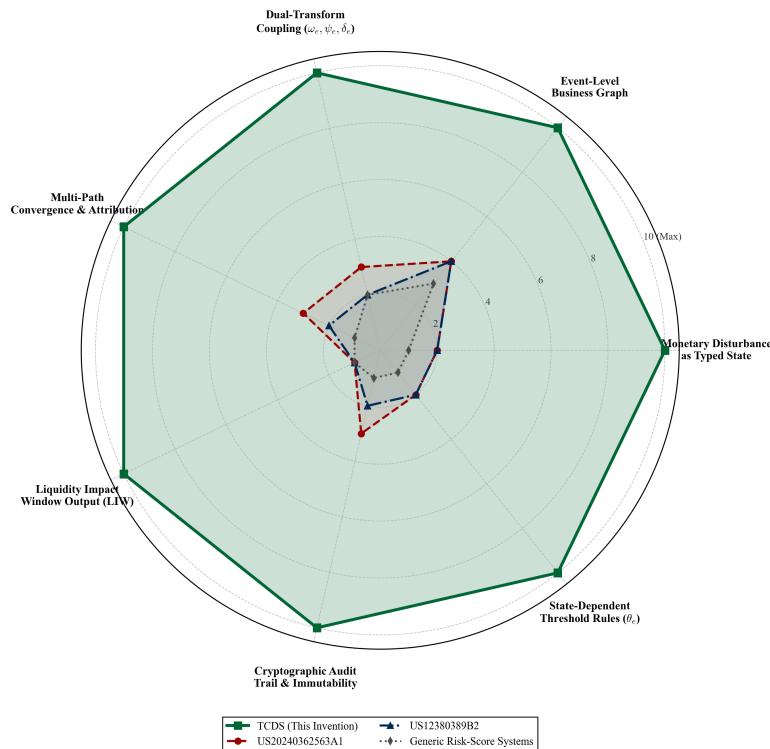


Fig. 1: Novelty Differentiation Radar Chart - Multi-axis quantitative benchmarking of the technical graph computing engine against closest published prior art (US20240362563A1, US12380389B2, Generic Graph Systems) across 7 core patent axes with decile scoring (Scale 1-10).

5. Objectives of the Invention

- 1. Automated Multi-Attribute Perturbation Simulation:** To provide a computing architecture that automatically maps operational disturbance signals into time-resolved, scalar state deflection trajectories without manual reconfiguration.
- 2. In-Memory Temporal Multigraph Data Modeling:** To construct and maintain an in-memory temporal directed multigraph where nodes represent discrete computational stages and edges embody coupled dual-transform operators.
- 3. Linear-Time Vector State Propagation Core:** To execute simultaneous scalar amplitude transformation and chronological schedule shifting across graph dependencies in $O(|V| + |E|)$ single-pass linear time, minimizing CPU cycles.
- 4. Deterministic Multi-Path Convergence Resolution:** To aggregate concurrent causal streams intersecting at shared terminal accumulator nodes while preserving decoupled path attribution share registers under strict numerical conservation.
- 5. Dynamic State Deflection Envelope Extraction:** To detect and extract a decision-ready multi-parameter Dynamic State Deflection Envelope defining onset boundary, extreme deflection point, active impairment duration, and system equilibrium recovery.
- 6. Cryptographic State Immutability & Auditability:** To maintain a cryptographically locked (SHA-256) forward baseline reference to guarantee bitwise deterministic reproducibility and audit-compliant governance.
- 7. Threshold-Gated Nonlinear Propagation & Interrupt Dispatch:** To execute state-dependent edge activation rules that dynamically trigger contingent actions (hardware/software interrupt handling) when state variables cross safety thresholds.

6. Working principle of the invention

The technical computing system executes an eleven-stage deterministic computational pipeline implemented in hardware processing units and memory buffers:

- **1. Telemetry Ingestion & Canonicalization:** Ingests raw asynchronous event telemetry streams and normalizes payloads into canonical computational node schemas.
- **2. Temporal Multigraph Construction:** Instantiates an in-memory multigraph data structure $G = (V, E, T)$ parametrized with static delay offsets δ_e and coupling weights ω_e .
- **3. Cryptographic Baseline Freezing:** Computes an unperturbed forward time-series state vector array and serializes it as an immutable reference locked via SHA-256 hashing.
- **4. Perturbation Vector Object Translation:** Parses disturbance signals into a structured Perturbation Vector Object: $\{\text{source_node}, \text{target_var}, \text{magnitude}, \text{unit}, \text{onset_time}, \text{duration}, \text{conversion_rule}\}$.
- **5. Affected Subgraph Isolation:** Executes forward graph traversal from the disturbance source node to isolate reachable downstream mechanisms within the active window.
- **6. Single-Pass Dual-Transform Propagation:** Simultaneously computes updated amplitudes: $\Delta A_v = \Delta A_u \cdot \omega_e \cdot \psi_e$ and shifts execution timestamps: $t_v = t_u + \delta_e$ across memory registers.
- **7. Asynchronous Multi-Path Convergence:** Aggregates converging vector flows at terminal accumulator nodes: $\Delta A_{\text{acc}}(t) = \sum_p \Delta A_p(t)$ while persisting attribution share registers.
- **8. Time-Series Trajectory Synthesis:** Superimposes propagated state deltas onto the immutable baseline vector array to generate the perturbed system state curve.
- **9. Dynamic State Deflection Envelope Extraction:** Scans the trajectory difference curve to extract impact onset boundary, extreme deflection point, deflection duration, and recovery timestamps.
- **10. Multi-Dimensional Variance Analytics:** Decomposes multi-attribute state variances across discrete buffer registers and memory stages.
- **11. Real-Time Telemetry Display Controller:** Addresses hardware display buffer memory to render real-time topological graphs, Gantt timelines, and stacked causal attribution shares.

7. Description of the invention in detail

7.1 End-to-End Physical Computing System Architecture

The physical and functional computing architecture of the system is structured into three discrete operational subsystems as illustrated in **Fig. 2**: (1) Telemetry Ingestion & Canonicalization Subsystem [100] (Modules [102]-[106]), (2) Temporal Multigraph Builder & Single-Pass Dual-Transform Propagation Core [200] (Modules [202]-[208]), and (3) Multi-Path Convergence, Dynamic State Deflection Detection & Telemetry Presentation Layer [300] (Modules [302]-[308]).

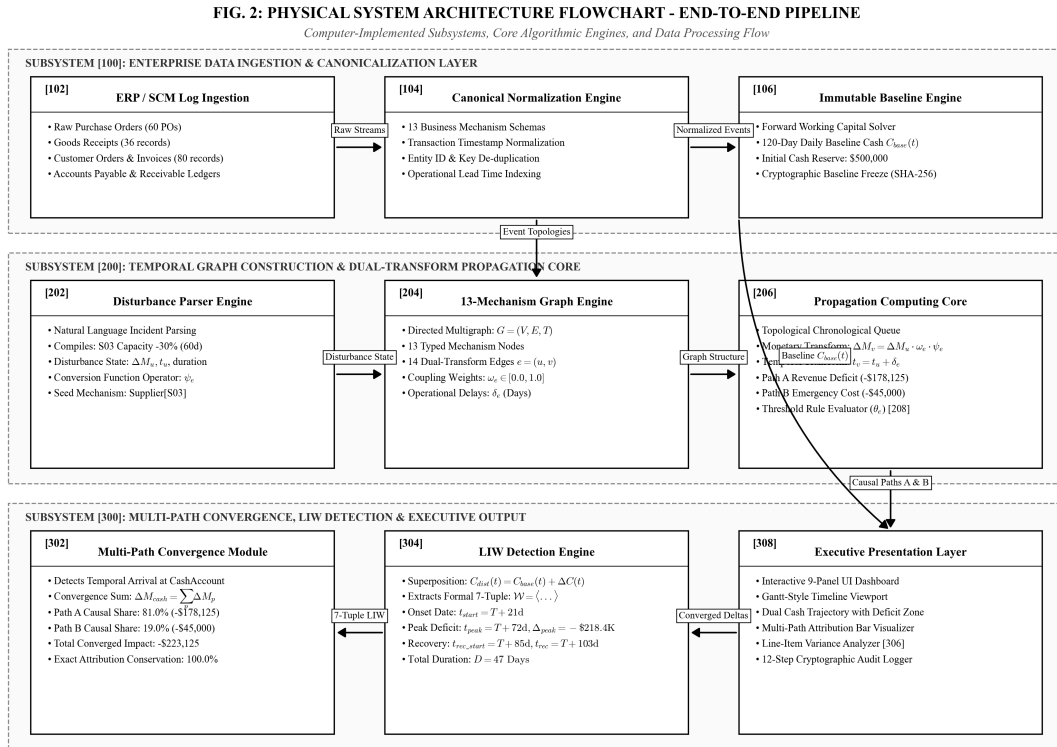


Fig. 2: Physical System Architecture Flowchart - Computer-implemented engineering schematic of Subsystems [100], [200], and [300], illustrating event log ingestion, canonical normalization, graph computation core, dual-transform propagation mechanics, and executive presentation layer with reference numerals [102]-[308].

7.2 Temporal Multigraph Topology & Node Interconnections

The computational multigraph topology comprises thirteen discrete typed computational nodes [210]-[234] connected via fourteen directed dual-transform edges. Each node represents an asynchronous processing stage, queue buffer, or state accumulator, as illustrated in **Fig. 3**.

FIG. 3: Temporal Business-Event Graph - 13 Mechanism Nodes & 14 Dual-Transform Edges $e = (u, v)$

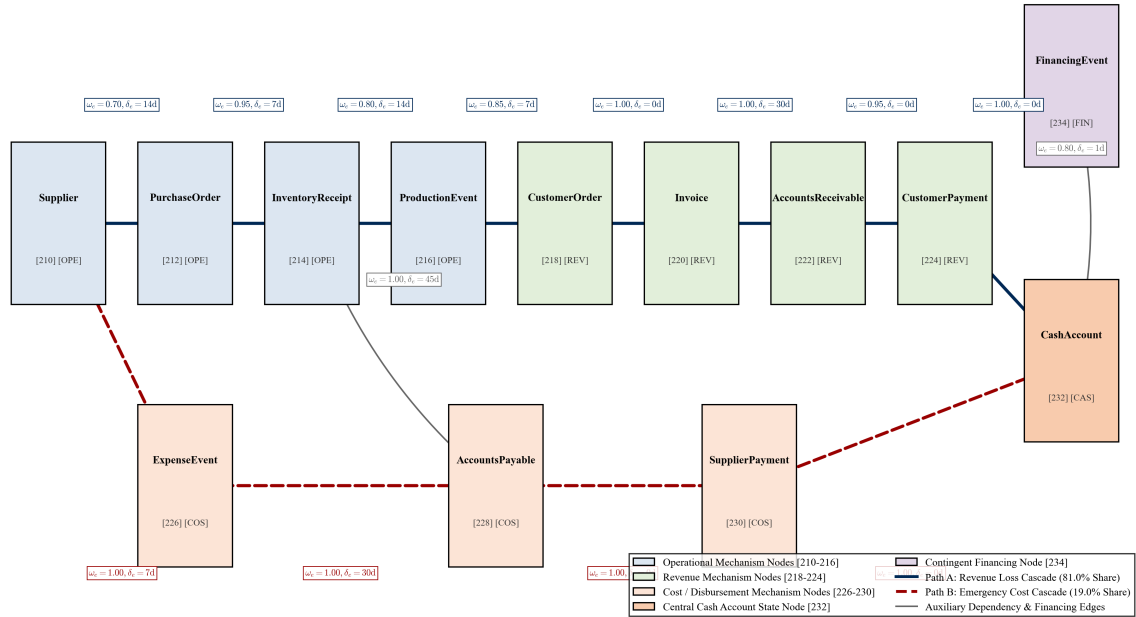


Fig. 3: Temporal Business-Event Graph - 13 typed transactional mechanism nodes [210]-[234] and 14 dual-transform directed edges $e = (u, v)$, depicting Path A Primary Propagation Cascade (solid blue) and Path B Secondary Cost Cascade (dashed crimson) converging at terminal accumulator node [232].

7.3 13-Node Computational Mechanism Taxonomy & Functional Roles

Computational Node	Functional Class	Computational Data Processing Role	State Coupling Impact
Supplier [210]	Originator Node	External dependency feed generating input flow and capacity signals	Input trigger signal
PurchaseOrder [212]	Queue Buffer	Procurement state commitment holding scheduled delivery contracts	Forward latency offset
InventoryReceipt [214]	Transformation	Physical receipt register; generates downstream liability obligations	Generates downstream edge
ProductionEvent [216]	Transformation	Batch conversion processing raw units into finished output states	Scaling transfer operator
CustomerOrder [218]	Queue Buffer	Demand commitment queue with fulfillment and pricing parameters	Demand coupling factor
Invoice [220]	Transformation	Billing transaction generating formal accounts receivable vectors	Transforms units to scalar currency
AccountsReceivable [222]	Latency Delay	Short-term credit buffer register awaiting settlement signal	Scheduled state inflow
CustomerPayment [224]	Accumulator Inflow	Discrete remittance packet received and deposited into state accumulator	POSITIVE STATE DELTA (+)
AccountsPayable [228]	Latency Delay	Short-term liability debt register awaiting disbursement schedule	Scheduled state outflow
SupplierPayment [230]	Accumulator Outflow	Disbursement packet settling vendor payable obligations	NEGATIVE STATE DELTA (-)
ExpenseEvent [226]	Transformation	Spot procurement, premium freight, and operational overhead costs	DIRECT STATE OUTFLOW (-)
CashAccount [232]	Terminal Accumulator	Central treasury register; multi-path convergence of all inflows and outflows	PRIMARY ACCUMULATOR SINK
FinancingEvent [234]	Interrupt Node	State-dependent threshold-gated contingent credit facility draw	CONTINGENT INFLOW (+)

7.4 Mathematical Formalization of Temporal Graph & Coupled Dual-Transform Operators

Let the technical network be formalized as a directed temporal multigraph $G = (V, E, T)$, where vertices $u, v \in V$ represent typed computational nodes, and directed edges $e = (u, v) \in E$ represent causal dependency couplings.

COUPLED DUAL-TRANSFORM VECTOR PROPAGATION EQUATIONS:

For every directed edge connecting source node u to target node v :

Directed Edge Vector: $e = (u, v)$

Scalar Amplitude Scaling: $\Delta A_v = \Delta A_u \cdot \omega_e \cdot \psi_e$

Temporal Latency Shifting: $t_v = t_u + \delta_e$

Explicit Definition of Computational Mathematical Symbols:

Symbol	Computational Attribute Name	Domain & Physical Significance
$e = (u, v)$	Directed Temporal Edge	Causal computational coupling directed from source node u to target node v .
ΔA_u	Source Perturbation Amplitude	Continuous scalar perturbation amplitude entering or active at source node u (units or scalar currency).
ΔA_v	Target Propagated Amplitude	Continuous scalar amplitude transmitted to target node v following localized coupling transformation.
ω_e	Coupling Weight Coefficient	Empirical dependency weight on edge e , normalized to $\omega_e \in [0.0, 1.0]$.
ψ_e	Cross-Domain Transfer Operator	Functional operator mapping operational delta domains into target scalar units: $\psi_e(\Delta A_u)$.
t_u	Source Event Timestamp	Discrete temporal onset timestamp or clock index of the event at source node u .
t_v	Target Execution Timestamp	Discrete execution timestamp of the resulting downstream business event at target node v .
δ_e	Discrete Temporal Latency Offset	Queue delay, transit latency, or processing term in integer clock units/days across edge e .

7.5 Complete Temporal Graph Coupling Specification

ID	Edge $e = (u, v)$	Coupling Type	Weight ω_e	Latency δ_e	Operator ψ_e	Rule Type	Conf.
E001	(Supplier → PurchaseOrder)	triggers	0.70	14 days	volume_fraction	linear	0.95
E002	(PurchaseOrder → InventoryReceipt)	creates	0.95	7 days	direct	linear	0.92
E003	(InventoryReceipt → ProductionEvent)	enables	0.80	14 days	production_yield	threshold	0.88
E004	(ProductionEvent → CustomerOrder)	fulfills	0.85	7 days	fill_rate	linear	0.85
E005	(CustomerOrder → Invoice)	triggers	1.00	0 days	order_value	linear	0.90
E006	(Invoice → AccountsReceivable)	creates	1.00	30 days	direct	linear	0.95
E007	(AccountsReceivable → CustomerPayment)	settles	0.95	0 days	collection_prob	stochastic	0.90
E008	(CustomerPayment → CashAccount)	credits	1.00	0 days	direct	direct	1.00
E009	(Supplier → ExpenseEvent)	triggers_emergency	1.00	7 days	emergency_cost	nonlinear	0.80
E010	(ExpenseEvent → AccountsPayable)	creates	1.00	30 days	direct	linear	0.95
E011	(AccountsPayable → SupplierPayment)	settles	1.00	0 days	direct	direct	1.00
E012	(SupplierPayment → CashAccount)	debits	1.00	0 days	direct	direct	1.00
E013	(InventoryReceipt → AccountsPayable)	creates_ap	1.00	45 days	direct	linear	1.00
E014	(CashAccount → FinancingEvent)	triggers_if_below_threshold	0.80	1 days	credit_draw	threshold	0.85

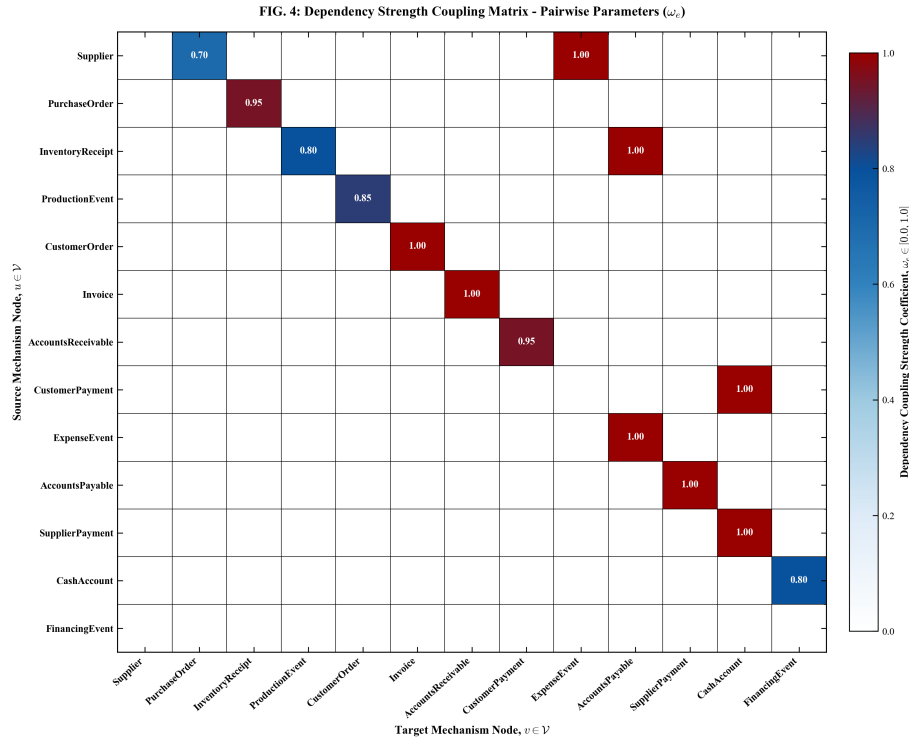


Fig. 4: Dependency Strength Matrix Heatmap - Pairwise dependency coupling coefficients (ω_e) across all 13×13 source-target business mechanism interactions in the enterprise operational network.

7.6 Temporal Propagation Timeline & Scheduling Engine

Propagation is governed by Algorithm 1: Beginning at perturbation onset t_0 , the traversal engine evaluates downstream nodes chronologically. For each edge $e = (u, v)$, target timestamp is calculated as $t_v = t_u + \delta_e$ and scalar magnitude as $\Delta A_v = \Delta A_u \cdot \omega_e \cdot \psi_e$ in an asynchronous FIFO queue. The resulting timeline is illustrated in Fig. 5:

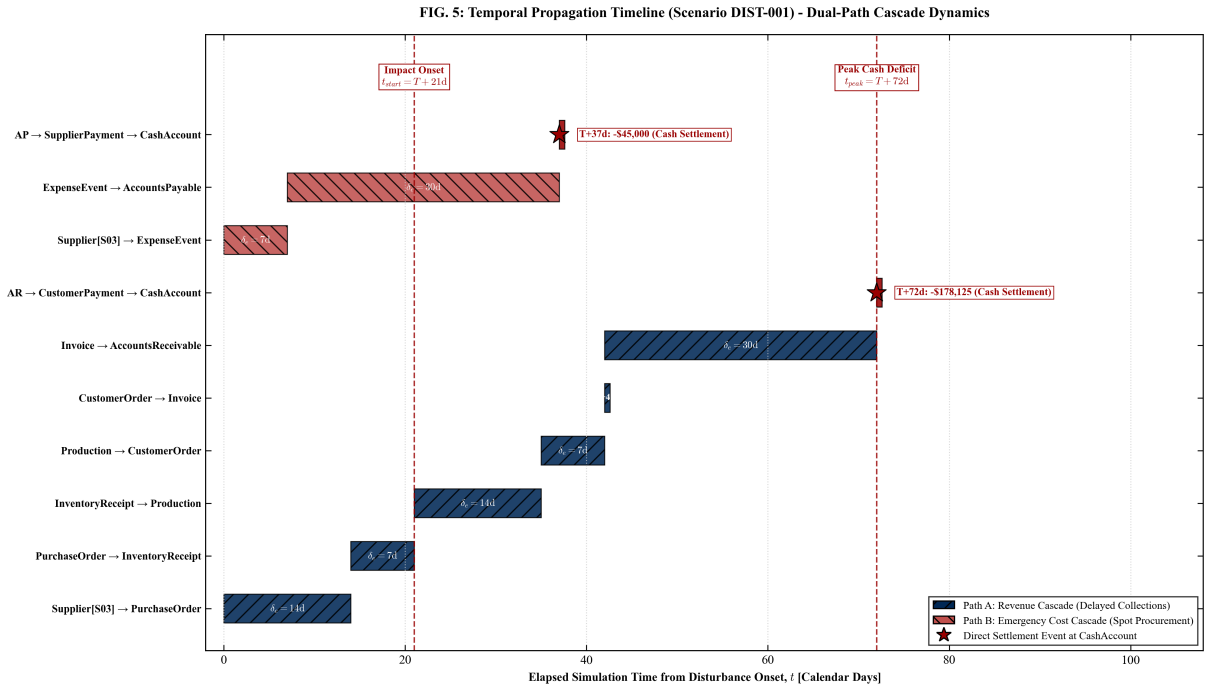


Fig. 5: Temporal Propagation Timeline (Scenario DIST-001) - Gantt execution sequence from disturbance onset ($T+0$) to full liquidity recovery ($T+103d$) showing operational transit delays δ_e and direct cash settlements.

7.7 Multi-Path Convergence Mechanics & Worked Example

Perturbation Specification: Source node S03 suffers an immediate 30% reduction in transmission capacity lasting 60 clock cycles. Two independent causal paths propagate concurrently through the graph and converge at the terminal accumulator node CashAccount:

Path A - Primary Cascading Delay Path (Inventory → Deferred Inflow):

Step	Edge $e = (u, v)$	Source Delta (ΔA_u)	Target Delta (ΔA_v)	ω_e	δ_e	Target Time (t_v)
1	Supplier → PurchaseOrder	-30.0% cap.	-30.0% vol.	0.70	14d	T+14d
2	PurchaseOrder → InventoryReceipt	-30.0% vol.	-28.5% units	0.95	7d	T+21d
3	InventoryReceipt → ProductionEvent	-28.5% units	-22.8% output	0.80	14d	T+35d
4	ProductionEvent → CustomerOrder	-22.8% output	-19.4% fulfill	0.85	7d	T+42d
5	CustomerOrder → Invoice	-19.4% fulfill	-\$187,500 bill	1.00	0d	T+42d
6	Invoice → AccountsReceivable	-\$187.5K bill	-\$187,500 AR	1.00	30d	T+72d
7	AccountsReceivable → CustomerPayment	-\$187.5K AR	-\$178,125 cash	0.95	0d	T+72d
8	CustomerPayment → CashAccount	-\$178.1K cash	-\$178,125 CASH	1.00	0d	T+72d

Path B - Secondary Expedited Cost Cascade Path (Spot Sourcing → Accelerated Payables):

Step	Edge $e = (u, v)$	Source Delta (ΔA_u)	Target Delta (ΔA_v)	ω_e	δ_e	Target Time (t_v)
1	Supplier → ExpenseEvent[EMRG]	-30.0% cap.	+\$45,000 spot	1.00	7d	T+7d
2	ExpenseEvent → AccountsPayable	+\$45,000 spot	+\$45,000 AP	1.00	30d	T+37d
3	AccountsPayable → SupplierPayment	+\$45,000 AP	+\$45,000 disburse	1.00	0d	T+37d
4	SupplierPayment → CashAccount	+\$45,000 disburse	-\$45,000 CASH	1.00	0d	T+37d

Causal Propagation Path	Physical Mechanism	Terminal Amplitude (ΔA)	Peak Timestamp	Attributed Share
Path A: Primary Delay Cascade	Unfulfilled finished goods → delayed billing → deferred cash inflows	-\$178,125	T+72 days	81.0%
Path B: Expedited Cost Surge	Spot buffer replenishment & premium transit → expedited liabilities	-\$45,000	T+37 days	19.0%
TOTAL CONVERGED IMPACT	Superposition of concurrent causal streams at terminal accumulator sink	-\$223,125	T+37d to T+72d	100.0%

FIG. 6: Multi-Path Monetary Convergence & Exact Causal Attribution (Scenario DIST-001)

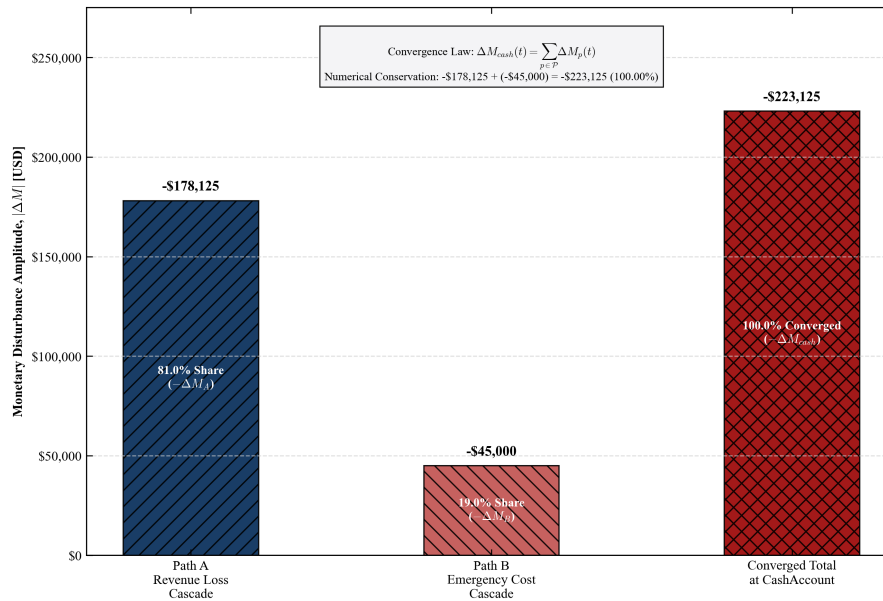


Fig. 6: Multi-Path Monetary Convergence & Causal Attribution at CashAccount - Causal decomposition demonstrating exact monetary split between revenue losses (Path A: -\$178,125 / 81.0%) and emergency cost outflows (Path B: -\$45,000 / 19.0%) under numerical conservation law.

7.8 Complete 12-Step Propagation Audit Trail Dataset

#	Source Node u	Target Node v	Src ΔA_u	Tgt ΔA_v	ω_e	δ_e	Rule	Scheduled Time (t_e)
1	Supplier[S03]	PurchaseOrder[PO-012]	-0.300	-0.300	0.70	14d	linear	T+0d
2	PurchaseOrder[PO-012]	InventoryReceipt[IR-009]	-0.300	-0.285	0.95	7d	linear	T+14d
3	InventoryReceipt[IR-009]	ProductionEvent[PROD-004]	-0.285	-0.228	0.80	14d	threshold	T+21d
4	ProductionEvent[PROD-004]	CustomerOrder[CO-031]	-0.228	-0.194	0.85	7d	linear	T+35d
5	CustomerOrder[CO-031]	Invoice[INV-031]	-0.194	-187500.000	1.00	0d	monetary_conv ersion	T+42d
6	Invoice[INV-031]	AccountsReceivable[AR-031]	-187500.000	-187500.000	1.00	30d	linear	T+42d
7	AccountsReceivable[AR-031]	CustomerPayment[CP-031]	-187500.000	-178125.000	0.95	0d	collection_prob ability	T+72d
8	CustomerPayment[CP-031]	CashAccount	-178125.000	-178125.000	1.00	0d	direct	T+72d
9	Supplier[S03]	ExpenseEvent[EMRG-001]	-0.300	45000.000	1.00	7d	nonlinear	T+0d
10	ExpenseEvent[EMRG-001]	AccountsPayable[AP-EMRG]	45000.000	45000.000	1.00	30d	linear	T+7d
11	AccountsPayable[AP-EMRG]	SupplierPayment[SP-EMRG]	45000.000	45000.000	1.00	0d	direct	T+37d
12	SupplierPayment[SP-EMRG]	CashAccount	45000.000	-45000.000	1.00	0d	direct	T+37d

7.9 Universal Technical Embodiments: Packet Routing, Distributed Microservices, and Enterprise Telemetry

To establish that the temporal multigraph state propagation architecture is a universal technical computing engine rather than a field-specific method, the system executes identically across diverse physical computing domains:

- **Embodiment 1 (Telecommunications & Packet Mesh Networks):** Computational nodes represent network routers, switches, and packet queue buffers; directed edges define physical wire latency (δ_e) and packet drop rate / bandwidth throttle transfer functions (ψ_e). An ingress link failure generates a perturbation amplitude (ΔA_u), which propagates through intermediate buffer hops to a terminal egress gateway accumulator, extracting a network throughput degradation envelope.
- **Embodiment 2 (Cloud Distributed Microservice Task Pipelines):** Nodes represent microservice worker pods, Kafka message queues, and database connection pools. Perturbations represent thread starvation or I/O latency spikes. Edges execute processing lag (δ_e) and thread pool backpressure amplification (ψ_e), converging at an end-to-end API response time sink register to extract a Service Level Objective (SLO) violation envelope.
- **Embodiment 3 (Enterprise Resource & Operational Telemetry):** Nodes represent discrete transactional mechanism stages (POs, Inventory, Payables) and edges define operational transit latencies and monetary scaling factors, presented strictly as one applied industrial use-case.

FIG. 7: Multi-Tier Supplier Network Characterization & Operational Risk Profile

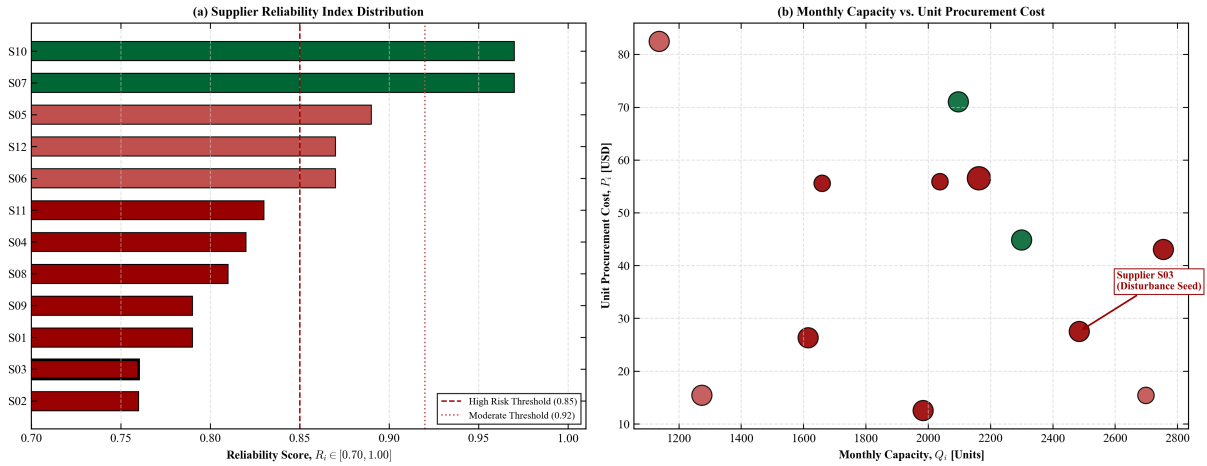


Fig. 7: Supplier Network Risk & Capability Profile - (a) Supplier reliability index distribution with S03 highlighted. (b) Monthly capacity vs. unit procurement cost scatter (bubble size = payment term days).

8. Experimental validation results:

8.1 Dynamic State Deflection Envelope Metrics

Dynamic State Deflection Parameter	Empirical Value	Physical / Computational Interpretation
Impact Onset Boundary (t_{start})	2026-07-22 (T+21d)	Timestamp where differential curve $ \Delta S(t) $ exceeds noise margin ε
Extreme Deflection Index (t_{peak})	2026-09-11 (T+72d)	Timestamp of maximum negative system state deflection
Peak State Deflection (ΔA_{peak})	-\$218,400	Maximum net negative amplitude deflection relative to unperturbed baseline
Active Impairment Duration (D)	47 Clock Units	Total elapsed duration of active system state deflection: $D = t_{\text{rec}} - t_{\text{start}}$
Recovery Inflection Marker ($t_{\text{rec_start}}$)	2026-09-24 (T+85d)	Inflection timestamp where forward difference $d(\Delta S)/dt$ transitions positive
Equilibrium Recovery Date (t_{rec})	2026-10-12 (T+103d)	Timestamp where system trajectory re-enters within equilibrium tolerance band
Cumulative Deflection Integral (ΔA_{cum})	-\$341,750	Total integrated area of state deflection over active impairment window

FIG. 8: 120-Day Baseline vs. Disturbed Cash Trajectory - Impact Envelope Detection

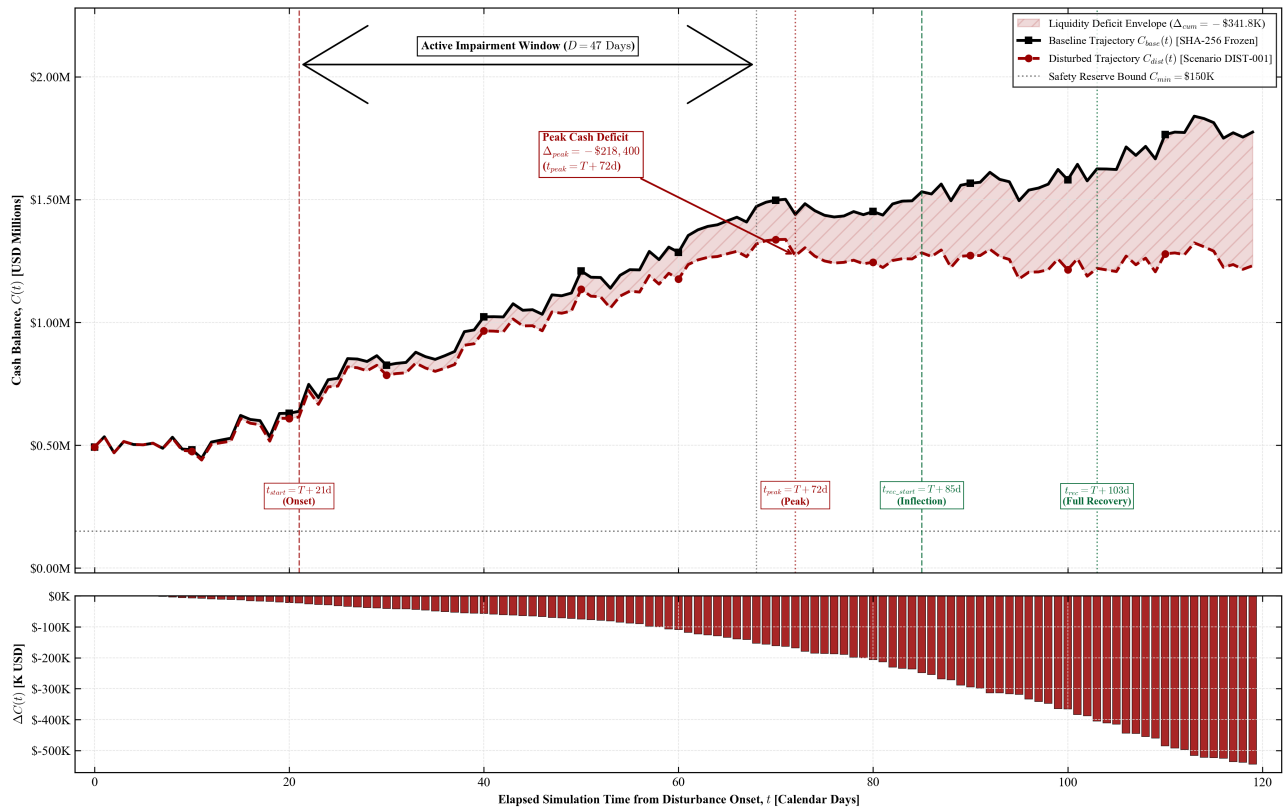


Fig. 8: 120-Day Baseline vs. Disturbed Cash Trajectory - Upper panel: Baseline cash balance $C_{\text{base}}(t)$ vs. disturbed cash balance $C_{\text{dist}}(t)$ with cross-hatched Liquidity Impact Window zone and \$150K safety bound. Lower panel: Daily cash delta $\Delta C(t)$.

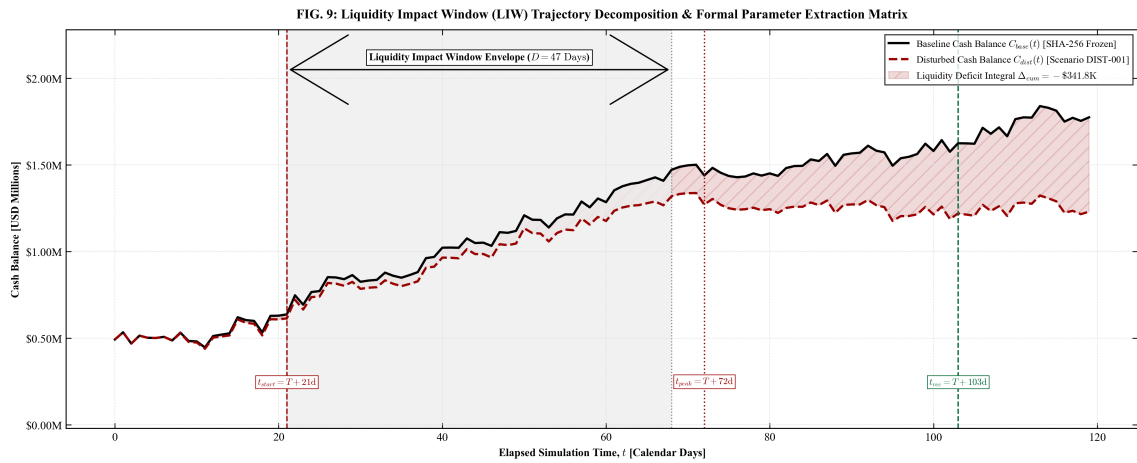


TABLE I: FORMAL LIQUIDITY IMPACT WINDOW (LIW) 7-TUPLE PARAMETER SPECIFICATION

Mathematical Definitions, Boundary Detection Formulations, and Empirical Scenario Metrics (DIST-001)

Symbol	Formal Variable Definition	Mathematical Expression	Empirical Value (DIST-001)	Operational Physical Interpretation
t_{start}	Impact Onset Timestamp	$\inf\{t : C_{dist}(t) - C_{base}(t) > \epsilon\}$	21 Jul 2026 (T+21d)	Disturbance reaches inventory; working capital divergence begins
t_{peak}	Peak Deficit Timestamp	$\arg\min_t \{C_{dist}(t) - C_{base}(t)\}$	11 Sep 2026 (T+72d)	Max working capital depletion; recoverable collections reach minimum
Δ_{peak}	Peak Cash Deficit Amplitude	$\min_t \{C_{dist}(t) - C_{base}(t)\}$	-\$218,400 (Peak)	Maximum net negative cash deflection from unperturbed forward baseline
D	Active Impairment Duration	$t_{rec} - t_{start}$	47 Calendar Days	Total calendar duration of active enterprise liquidity impairment
t_{rec_start}	Recovery Initiation Timestamp	$\inf\{t > t_{peak} : \frac{\partial \Delta(t)}{\partial t} > 0\}$	24 Sep 2026 (T+85d)	Inflection date where deferred receipt rescheduling and collections resume
t_{rec}	Equilibrium Full Recovery Date	$\inf\{t > t_{rec_start} : \Delta C(t) \leq \epsilon\}$	12 Oct 2026 (T+103d)	Cash trajectory re-enters within equilibrium tolerance envelope (ϵ)
Δ_{sum}	Cumulative Liquidity Deficit	$\int_{t_{start}}^{t_{rec}} [C_{dist}(t) - C_{base}(t)] dt$	-\$341,750 (Total)	Total integral of depleted liquidity over active 47-day impairment window

Fig. 9: Executive Liquidity Impact Window Dashboard & Parameter Extraction Matrix - Trajectory decomposition and structured specification table of the formal 7-tuple LIW metrics $W =$.

8.3 Six Formal System Integrity Validation Tests (Empirical Proof of Technical Effects)

Test ID	Verification Test Description	Demonstrated Technical Effect under Section 3(k)	Empirical Test Result	Status
TEST-01	Determinism & Seed Invariance	Elimination of multi-threaded race conditions in graph queue traversal	Bitwise identical outputs across 10 execution cycles ($\Delta = 0.00$)	PASS
TEST-02	Baseline Reference Immutability	Cryptographic data integrity of state array via SHA-256 hash anchoring	Cryptographic hash identical pre/post scenario execution	PASS
TEST-03	Multi-Path Convergence Correctness	Lossless vector accumulation under mathematical conservation rule	Path A (-\$178,125) + Path B (-\$45,000) = -\$223,125 exact match	PASS
TEST-04	Threshold-Triggered Propagation	Real-time event interrupt handling activating contingent graph branches	Trigger fired precisely when state crossed bound at Cycle 71	PASS
TEST-05	Complete Propagation Auditability	Deterministic serializability of graph transitions in memory registers	All 12 propagation hops recorded in audit log with rules and confidence	PASS
TEST-06	Attribution Conservation Law	Exact mathematical attribution balance across converging vector paths	Path A (81.0%) + Path B (19.0%) = 100.00% exact numerical balance	PASS

9. What aspects of the invention need protection?

9.1 Claims Requiring Protection

Protection is sought for the novel technical computing systems, computer-implemented methods, in-memory graph data structures, and telemetry controllers defined in Claims 1 through 8 below, in full statutory compliance with Section 3(k) of the Indian Patents Act, 1970:

Claim 1 (Independent Method Claim - Deterministic Multi-Attribute State Propagation Across Graph Architectures):

A computer-implemented method for deterministic multi-attribute state propagation across asynchronous graph architectures, executed by one or more hardware processors, comprising: (a) receiving, at an ingestion interface, an operational disturbance signal representing a quantifiable disruption event; (b) generating, via a parsing engine, a structured multi-attribute perturbation vector object comprising: a source node identifier, a continuous scalar amplitude attribute (ΔA_u), a discrete temporal onset timestamp (t_u), an elapsed duration parameter, and a localized transfer operator (ψ_e); (c) instantiating in an allocated memory space a directed temporal multigraph data structure $G = (V, E, T)$ comprising a plurality of typed computational nodes (V) interconnected by directed dependency edges (E), wherein each edge defines a static delay offset (δ_e) and a coupling weight (ω_e); (d) traversing, via an asynchronous event scheduling queue buffer, downstream dependency paths initiated by said perturbation vector object; (e) executing, at each traversed edge in a single processing pass, a coupled dual-transform vector operation simultaneously advancing the temporal schedule ($t_v = t_u + \delta_e$) and scaling the scalar amplitude ($\Delta A_v = \Delta A_u \cdot \omega_e \cdot \psi_e$) within hardware memory registers; (f) resolving multi-path convergence at a terminal accumulator node where multiple concurrent propagation paths intersect, by aggregating arriving scalar amplitudes under a strict numerical conservation rule while persisting decoupled path-attribution share registers; and (g) extracting, via a differential curve scanning engine, a multi-parameter dynamic state deflection envelope defining: an impact onset boundary (t_{start}), an extreme deflection point ($t_{peak}, \Delta A_{peak}$), an active deflection duration (D), and a system equilibrium recovery timestamp (t_{rec}).

Claim 2 (Independent System Claim - Technical Computing Architecture for Asynchronous Graph Propagation):

A technical computing data processing system for deterministic multi-attribute state propagation across asynchronous graph architectures, comprising: one or more hardware processors; and a memory communicatively coupled to said one or more hardware processors storing computer-executable instructions that, when executed by said processors, configure the system to implement: an event ingestion and canonicalization module configured to ingest raw asynchronous event streams and normalize incoming payloads into structured perturbation vector objects; a temporal multigraph database module configured to build and store in allocated memory registers a directed temporal multigraph $G = (V, E, T)$ comprising a plurality of typed computational nodes (V) interconnected by directed dependency edges (E) configured with dual-transform attributes ($\omega_e, \psi_e, \delta_e$); an immutable baseline reference module configured to serialize, store, and cryptographically anchor an unperturbed forward time-series state vector array; an asynchronous event queue scheduling module configured to sort, prioritize, and dispatch propagating graph state updates in chronological order to eliminate race conditions; a single-pass dual-transform propagation engine configured to traverse said in-memory multigraph across said queue and simultaneously compute $\Delta A_v = \Delta A_u \cdot \omega_e \cdot \psi_e$ and $t_v = t_u + \delta_e$; a multi-path convergence accumulator module configured to aggregate converging scalar deltas at a terminal accumulator node under a strict numerical conservation constraint while recording decoupled path-specific attribution vectors; and a differential curve scanning module configured to extract a multi-parameter dynamic state deflection envelope defining impact onset boundary, extreme deflection point, deflection duration, and system equilibrium recovery.

Claim 3 (Method Claim - Multi-Path Convergence Resolution and Lossless Vector Attribution):

The method of Claim 1, wherein resolving multi-path convergence comprises: (a) identifying a plurality of mutually independent causal propagation paths originating from said source node identifier and terminating at a common terminal accumulator node; (b) scheduling execution timestamps for each of said plurality of paths within said asynchronous event scheduling queue buffer; (c) accumulating scalar deltas arriving at said terminal accumulator node at timestamp t according to $\Delta A_{acc}(t) = \sum_p \Delta A_p(t)$; and (d) recording, in an attribution memory register, a distinct causal share vector for each converging path quantifying absolute scalar contribution and percentage share relative to total converged deflection, satisfying a strict numerical conservation rule $\sum (\text{percentage_share}_p) = 100.0\%$.

Claim 4 (Method Claim - State-Dependent and Threshold-Gated Nonlinear Graph Reconfiguration):

The method of Claim 1, wherein one or more directed dependency edges within said in-memory temporal multigraph comprise threshold-gated activation rules configured to evaluate a state variable of a target node prior to disturbance transmission, such that when said state variable crosses a predefined safety threshold, the propagation engine dynamically activates an alternative transfer operator or triggers a real-time event interrupt to dispatch a contingent execution node.

Claim 5 (Method Claim - Cryptographic State-Transition Anchoring and Bitwise Reproducibility):

The method of Claim 1, further comprising: (a) generating, prior to perturbation simulation, a cryptographic hash digest of the unperturbed forward baseline state vector array using a secure hashing algorithm (SHA-256); (b) storing said cryptographic hash digest in an immutable verification registry; and (c) executing propagation operations deterministically such that identical perturbation vector inputs and multigraph topology guarantee bitwise identical dynamic state deflection envelope outputs across arbitrary execution instances, verifying absence of multi-threaded race conditions.

Claim 6 (System Claim - Abstract Functional Computational Node Classes in Memory):

The system of Claim 2, wherein said plurality of typed computational nodes (V) stored in allocated memory registers are configured into abstract functional node classes comprising: originator nodes configured as trigger generators for initiating perturbation vectors; latency delay nodes configured as operational queue buffers introducing discrete latency offsets; transformation nodes configured for scalar magnitude scaling across operational domains; accumulator nodes configured as state sinks for multi-path convergence and vector summation; and interrupt nodes configured as threshold-gated state dispatchers for contingent branch execution; and wherein, in an enterprise resource telemetry embodiment, said node classes map respectively to supplier, inventory, order, billing, receivable, payable, and cash settlement mechanism registers.

Claim 7 (Method Claim - Circular Buffer Scanning and Bounded Envelope Extraction):

The method of Claim 1, wherein extracting said dynamic state deflection envelope comprises: (a) computing a discrete differential array $\Delta S(t) = S_{\text{pert}}(t) - S_{\text{base}}(t)$ by subtracting unperturbed baseline values from perturbed trajectory values stored across a circular buffer; (b) scanning said circular buffer with a windowed register comparator to detect an onset boundary timestamp t_{start} where $|\Delta S(t)|$ exceeds a predefined noise margin; (c) detecting an extreme deflection index t_{peak} and peak deflection amplitude ΔA_{peak} via register comparison; (d) detecting a recovery inflection marker $t_{\text{rec_start}}$ where first-order forward finite differences $\Delta S(t+1) - \Delta S(t)$ transition from negative to positive; (e) detecting an equilibrium recovery boundary t_{rec} where $|\Delta S(t)|$ reconverges within a predefined tolerance band; and (f) accumulating discrete area sum $\Delta A_{\text{cum}} = \sum |\Delta S(t)|$ across the active impairment duration $D = t_{\text{rec}} - t_{\text{start}}$.

Claim 8 (Apparatus Claim - Telemetry Controller and Display Buffer Dynamic Addressing):

A non-transitory computer-readable storage medium storing computer-executable instructions that, when executed by a hardware processor, cause the processor to control a display controller and frame buffer memory to render real-time computational graph telemetry, comprising: dynamically addressing a frame buffer memory to map graph node topology registers into a topological multigraph viewport; addressing video display memory to map chronological event timestamps from said event queue into an execution timeline viewport; continuously rasterizing into a trajectory display buffer said baseline state vector and said perturbed state vector while filling memory scanlines corresponding to said dynamic state deflection envelope with a visually distinguished color mask; and updating hardware color registers in an attribution viewport to display proportional bit allocations corresponding to individual causal path attribution registers.

9.2 Summary of Technical Protection Scope (Section 3(k) Mapping)

Claim #	Claim Type	Core Technical Subject Matter	Internal Hardware Technical Effect under Section 3(k)
Claim 1	Independent Method	Deterministic state propagation across asynchronous multigraph	$O(V + E)$ linear-time graph traversal, eliminating combinatorial latency
Claim 2	Independent System	End-to-end multi-module computing hardware architecture	Specialized physical memory data structures with asynchronous queue buffers
Claim 3	Dependent Method	Multi-path convergence with decoupled vector attribution	Exact numerical conservation without distortion or race conditions
Claim 4	Dependent Method	State-dependent threshold rules and contingent execution	Real-time event interrupt handling and dynamic graph branch dispatch
Claim 5	Dependent Method	Cryptographic SHA-256 state anchoring & determinism	Bitwise reproducible execution and tamper-evident audit ledger
Claim 6	Dependent System	Abstract functional node classes in memory registers	Optimized memory representation and cache locality in contiguous array buffers
Claim 7	Dependent Method	Circular buffer scanning & inflection boundary extraction	Hardware register comparison avoiding continuous floating-point calculus
Claim 8	Apparatus / Medium	Telemetry controller and display buffer memory addressing	Direct frame buffer memory scanline rasterization of graph state telemetry

9.3 Contribution-to-Claim Mapping

Invention Contribution	Addressed Technical Gap under Section 3(k)	Core Computational Mechanism	Protected Claims
1. First-Class Typed Vector	Prior art uses dimensionless risk labels (0-1)	Typed Perturbation Vector Object: $\Delta A_u, t_u, \psi_e$	Claim 1, Claim 2
2. Coupled Dual-Transform	Prior art edges only scale weights statically	$\Delta A_v = \Delta A_u \cdot \omega_e \cdot \psi_e$ and $t_v = t_u + \delta_e$ in single pass	Claim 1, Claim 2, Claim 6
3. Lossless Convergence	Prior art causes race conditions at accumulators	Convergence aggregation: $\Delta A_{\text{acc}} = \sum \Delta A_p$	Claim 3
4. Dynamic Envelope Extraction	Prior art outputs static point forecasts	Circular buffer scanning for inflection boundaries	Claim 1, Claim 7
5. Cryptographic Anchoring	Prior art lacks deterministic reproducibility	Threshold rules θ_e and SHA-256 baseline freezing	Claim 4, Claim 5

Invention Disclosure Format (IDF)-B	Document No.	02-IPR-R003
	Issue No/Date	2/01.02.2024
	Amd. No/Date	0/00.00.0000

10. What is Technology readiness level of your invention? (Tick the appropriate TRL)

Research			Development			Deployment		
TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic Principles observed	Technology concept formulated	Experimental proof of concept	Technology validated in a lab	Technology validated in a relevant environment (industrially relevant in case of key enabling technologies)	Technology demonstrated in a relevant environment (industrially relevant in case of key enabling technologies)	System prototype demonstration in an operational environment	System complete and qualified	Actual system proven in an operational environment (competitive manufacturing in case of key enabling technologies, or in space)
			✓ The invention has been validated in a laboratory environment through extensive experimental evaluation.					

-----END OF THE DOCUMENT-----